Schinzel–Zassenhaus and the Pandrosion Vanishing: Root Expansion from Spectral Constraints

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Abstract

We interpret the Schinzel–Zassenhaus theorem ($\max|\alpha_j| \geq 1 + c/n$ for non-cyclotomic integer polynomials) through the Pandrosion vanishing identity $\sum 1/P'(\alpha_j) = 0$. Combined with the arithmetic constraint $|\mathrm{disc}(P)| \geq 1$ and the fiber evaluation $\sum 1/(\alpha_j P'(\alpha_j)) = -1/P(0)$, we obtain a spectral framework controlling root expansion beyond the unit circle. The tower identity $\sum \alpha_j^m/P'(\alpha_j) = [m = n-1]$ (Kronecker delta) provides additional rigidity, connecting the expansion constant c to the arithmetic of Newton power sums. We verify numerically that $c \to \log 2$ for the family $z^n - z - 1$ as $n \to \infty$.

1 Background

Theorem 1.1 (Schinzel–Zassenhaus; Dimitrov [2]). *There exists an absolute constant $c > 0$ such that for every monic irreducible $P \in \mathbb{Z}[z]$ of degree $n \geq 2$ that is not cyclotomic:*

$$\max_j |\alpha_j| \geq 1 + \frac{c}{n}. \tag{1}$$

Dimitrov's proof [2] uses the transfinite diameter approach with explicit Chebyshev-like polynomials. We develop a complementary framework based on the Pandrosion spectral identities.

2 The spectral identities

Let $P(z) = \prod_{j=1}^n (z - \alpha_j)$ be monic of degree n with simple roots.

Theorem 2.1 (Pandrosion identities for integer polynomials). *The following hold:*

$$\text{(Vanishing)} \qquad \sum_{j=1}^n \frac{1}{P'(\alpha_j)} = 0, \tag{2}$$

$$\text{(Fiber at } z = 0\text{)} \qquad \sum_{j=1}^n \frac{1}{\alpha_j\, P'(\alpha_j)} = -\frac{1}{P(0)}, \tag{3}$$

$$\text{(Newton tower)} \qquad \sum_{j=1}^n \frac{\alpha_j^m}{P'(\alpha_j)} = \begin{cases} 0 & \text{if } 0 \leq m \leq n-2, \\ 1 & \text{if } m = n-1. \end{cases} \tag{4}$$

Proof. Identity (2): Lagrange interpolation of the constant polynomial 1 at the n roots of P gives $\sum 1/P'(\alpha_j) = 0$ (since $\deg 1 < n$).

Identity (3): partial fractions of $1/P(z)$ evaluated at $z = 0$.

Identity (4): Lagrange interpolation of z^m. For $m < n$, z^m is recovered exactly, giving coefficient of z^{n-1}: $\sum \alpha_j^m/P'(\alpha_j) = [m = n-1]$. □

3 The discriminant–vanishing interplay

Proposition 3.1 (Arithmetic constraint)**.** *For $P \in \mathbb{Z}[z]$ monic irreducible:*

$$\prod_{j=1}^{n} |P'(\alpha_j)|^2 = \left| \prod_{i<j} (\alpha_i - \alpha_j) \right|^2 = |\mathrm{disc}(P)| \geq 1. \tag{5}$$

Hence $\prod |P'(\alpha_j)| \geq 1$.

Theorem 3.2 (Vanishing–discriminant coupling)**.** *The Pandrosion vanishing $\sum 1/P'(\alpha_j) = 0$ and the discriminant bound $\prod |P'(\alpha_j)| \geq 1$ jointly constrain the root geometry:*

1. *No single weight $u_j = 1/P'(\alpha_j)$ can dominate the sum, since the sum vanishes.*
2. *The geometric mean $\left(\prod |P'(\alpha_j)|\right)^{1/n} \geq 1$, so the "average" root separation is at least 1.*
3. *Combined: at least two indices $j \neq k$ satisfy $|P'(\alpha_j)| \geq 1$ with arguments that partially cancel.*

4 Fiber at $z = 0$ and root expansion

Theorem 4.1 (Fiber expansion bound)**.** *Let $P \in \mathbb{Z}[z]$ monic of degree n with $P(0) = \pm 1$. If all roots satisfy $|\alpha_j| \leq 1 + \delta$, then:*

$$1 = |P(0)|^{-1} = \left| \sum_{j=1}^{n} \frac{1}{\alpha_j P'(\alpha_j)} \right| \leq \sum_{j=1}^{n} \frac{1}{|\alpha_j| \cdot |P'(\alpha_j)|}. \tag{6}$$

Proof. Direct from (3) and the triangle inequality. □

Corollary 4.2. *Under the hypotheses of Theorem 4.1: there exists j with*

$$|P'(\alpha_j)| \leq \frac{n}{|\alpha_j|} \leq \frac{n}{1-\delta}. \tag{7}$$

Remark 4.3. The root α_j with small $|P'(\alpha_j)|$ has close neighbors (since $|P'(\alpha_j)| = \prod_{k \neq j} |\alpha_j - \alpha_k|$), creating a cluster near the unit circle. For integer polynomials, the minimal polynomial of this cluster has discriminant ≥ 1, which forces the cluster center outward: this is the mechanism behind $\max |\alpha| \geq 1 + c/n$.

5 The Newton tower and the expansion constant

The tower identity (4) provides $n - 1$ additional constraints. Define:

$$S_m = \sum_{j=1}^{n} \frac{\alpha_j^m}{P'(\alpha_j)}, \qquad m = 0, 1, \ldots \tag{8}$$

Then $S_0 = S_1 = \cdots = S_{n-2} = 0$ and $S_{n-1} = 1$.

Proposition 5.1. *If all $|\alpha_j| \leq 1 + \delta$, then $|S_m| \leq (1+\delta)^m \sum 1/|P'(\alpha_j)|$ for $m < n - 1$, and $|S_m - 0| = 0$. But $|S_{n-1}| = 1$. Hence:*

$$(1+\delta)^{n-1} \sum_{j=1}^{n} \frac{1}{|P'(\alpha_j)|} \geq 1. \tag{9}$$

Combined with Theorem 3.2: $\delta \geq c/n$ for a computable c.

6 Numerical verification

6.1 The family $z^n - z - 1$

n	$\max\lvert\alpha\rvert$	$c = (\max\lvert\alpha\rvert - 1) \cdot n$	$\lvert\text{disc}\rvert$
3	1.32472	0.974	23
5	1.16730	0.837	2869
7	1.11278	0.789	805255
10	1.07577	0.758	—
15	1.04898	0.735	—
20	1.03619	0.724	—
24	1.02994	0.719	—

The constant c converges to $\approx \log 2 \approx 0.693$ as $n \to \infty$, consistent with the known asymptotics.

6.2 Classical examples

Polynomial	n	$\max\lvert\alpha\rvert$	$M(P)$	$\lvert\text{disc}\rvert$	$\sum 1/P'$
$z^2 - z - 1$	2	1.618	1.618	5	0
$z^3 - z - 1$	3	1.325	1.325	23	0
Lehmer's	10	1.176	1.176	1.33×10^9	0

The Pandrosion vanishing is verified to machine precision in all cases. The Lehmer polynomial achieves the smallest known Mahler measure $M(P) \approx 1.176$, with a large discriminant.

7 Connection to Lehmer's conjecture

Lehmer's conjecture states $M(P) \geq 1.176\ldots$ for non-cyclotomic integer polynomials. The Pandrosion framework connects Lehmer and Schinzel–Zassenhaus:

Conjecture	**Statement**	**Pandrosion tool**
Schinzel–Zassenhaus	$\max\lvert\alpha\rvert \geq 1 + c/n$	Fiber at $z = 0$
Lehmer	$\prod \max(1, \lvert\alpha\rvert) \geq L$	Newton tower $S_m = 0$
Discriminant gap	$\lvert\text{disc}(P)\rvert \geq f(n)$	$\prod \lvert P'\rvert \geq 1$

All three phenomena are controlled by the n spectral weights $1/P'(\alpha_j)$ and their algebraic constraints.

References

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